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1 What is Explainable AI (XAI)?

Explainable Artificial Intelligence (XAI) aims to make Machine Learning (ML) model predictions more understandable and support appropriate trust in these models [1]. It involves developing methods to explain model behaviour and communicating the resulting explanations in ways that address the needs of different stakeholders, including engineers, end users, and regulators [2].

Transparency, interpretability, and accountability are also important concepts in the broader context of trustworthy AI [2]. Transparency refers to the availability of relevant information about a system's design, operation, limitations, and decision-making processes. Interpretability concerns the extent to which humans can understand a model's outputs and how they relate to the underlying input data and context. Accountability requires the individuals and organizations responsible for developing, deploying, and using AI systems to justify their decisions and take responsibility for the consequences of their use [3]. These considerations are particularly important in high-stakes domains, such as healthcare and aviation.

2 Overview of XAI and Key Challenges

XAI aims to explain model predictions and support decision-making processes. It can help identify and investigate unexpected model behaviour, improve modelling and data collection processes, and assess whether a model meets the required criteria. Explanations can also help stakeholders understand the reasons behind model predictions [2].

An important challenge in XAI is determining what constitutes a good explanation. Explanations should be sufficiently complete, accurate, meaningful, and consistent for their intended purpose. However, achieving all these properties simultaneously can be difficult. In particular, a trade-off may arise between the accuracy of an explanation and its interpretability.

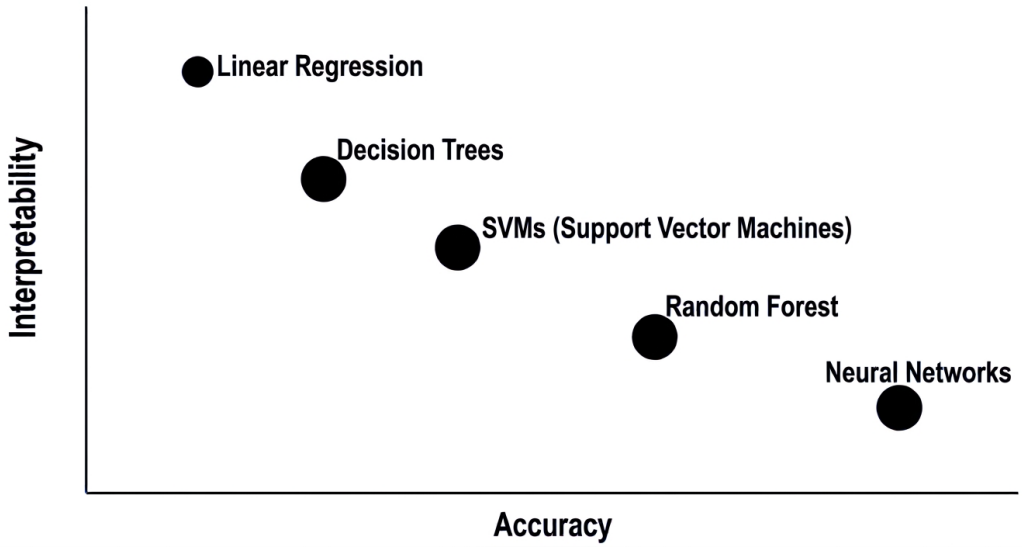


Figure 1: Illustration of the trade-off between predictive accuracy and interpretability. Adapted from [3].

As illustrated in Fig. 1, increasing predictive accuracy may be associated with decreasing interpretability as model complexity increases. For example, linear regression models are generally easier to interpret than complex neural networks, which can capture more complex patterns but are often harder to understand. This motivates the development of XAI techniques that help explain the predictions of complex models while maintaining their predictive performance.

3 Categorization of XAI methods

XAI methods can be classified according to different characteristics, including the scope of the explanation (global or local), their dependence on the underlying model (model-specific or model-agnostic), and the degree of access to the model's internal mechanisms (white-box or black-box).

3.1 White-Box vs. Black-Box

White-box models are transparent and comparatively easy to understand due to their internal structure. These models are often intrinsically interpretable, meaning their predictions can be understood without additional explanation methods. Examples include linear models, decision trees, and generalized additive models (GAMs) [3]. In contrast, black-box models are generally difficult to interpret due to the complexity of their internal mechanisms [3]. Neural networks (NNs), particularly deep neural networks, are common examples. *Post-hoc* methods are applied after model training to explain predictions and provide insights into model behaviour.

3.2 Model-Specific vs. Model-Agnostic

Model-specific techniques are designed for particular model architectures and exploit their internal structure to explain predictions [1, 3]. Examples include techniques developed for linear models and decision trees. Model-agnostic techniques can be applied to different ML models without requiring access to their internal structure. They explain predictions by analyzing the relationship between input features and model outputs [1, 3]. A well-known example is Local Interpretable Model-Agnostic Explanations (LIME).

3.3 Global vs. Local Interpretation

Global interpretation provides an overall understanding of a model's behaviour, whereas local interpretation focuses on individual predictions to explain why a particular output was generated [1, 3]. Examples include Partial Dependence Plots (PDPs) for global interpretation and LIME for local interpretation.

4 Final Discussion

This note provides an overview of XAI and its key concepts. For further reading, Ref. [1] provides the main theoretical background, complemented by the videos in Refs. [2, 3]. XAI techniques can be applied to different types of data, including tabular data, images, and graph structures. The following investigations will focus on XAI techniques for tabular and image data.

Appendix A

These notes are available on GitHub: `https://github.com/LucasFQM`. Readers are welcome to share comments, suggest corrections, and contribute ideas to improve and expand this material.

References

[1] Rudresh Dwivedi, Devam Dave, Het Naik, Smiti Singhal, Rana Omer, Pankesh Patel, Bin Qian, Zhenyu Wen, Tejal Shah, Graham Morgan, and Rajiv Ranjan. 2023. Explainable AI (XAI): Core Ideas, Techniques, and Solutions. *ACM Comput. Surv.* 55, 9, Article 194 (September 2023), 33 pages. https://doi.org/10.1145/3561048

[2] ML Tech Talks. Introduction to Explainable AI. *YouTube.* `https://youtu.be/6xePkn3-LME`

[3] Lecture 2 - Overview of Explainable AI (XAI) | Categories, Data Types and Methods. *YouTube.* `https://www.youtube.com/watch?v=-cRzYyTTjzY`